MATH60633A – TP2

David Ardia

Important information

- This assignment is worth 20% of the final grade.
- This work must be done in teams of 3-4 people; fill out the peer evaluation form.
- Send me an e-mail at david.ardia@hec.ca as soon as you have created your team.
- This assignment cannot be done using ChatGPT.
- I will only answer questions asked on the forum.
- Your Rmarkdown or R notebook file must be deposited before the date set for this purpose. Any delay will incur a 5% penalty per hour.
- Clearly document all your R code. It must be executable.
- Put all your files in a single .zip folder and name it `name1_name2.zip`. A single file per team must be submitted.
- The total number of points is 100.

Context

You work as a quantitative analyst for a large investment bank. You and your team are responsible for challenging the models used by traders and risk managers. You work with R and love reproducible research. All your files are written in Rmarkdown or R notebook.

You can watch the introductory video on Rmarkdown to help you build properly the R file.

Learning objectives

Content (scientific rigor, concepts, creativity, clarity) [10 points]

- Choose the right tools.
- Implement the steps correctly.
- Come up with innovative solutions.
- Report results with clarity.

Form (coding, collaboration, presentation) [20 points]

- Build RStudio project with proper folder structure
- Use Rmarkdown/nootebook file to reproduce your results.
- Program with state-of-the-art coding standards.

Description

The objective is to implement (part of) the risk management framework for estimating the risk of a book of European call options by taking into account the risk drivers such as underlying and implied volatility.

Instructions

Data

1. Load the database *Market*. Identify the price of the S&P 500, the VIX index, the term structure of interest rates (current and past), and the traded options (calls and puts).

Pricing of a portfolio of options [5 points]

1. Assume the following book of European call options: 1x strike $K = 1600$ maturity 20 days, 1x strike $K = 1650$ maturity 20 days, 1x strike $K = 1750$ maturity 40 days and 1x strike $K = 1800$ maturity 40 days.
2. Find the price of this book given the latest underlying price and the latest implied volatility (take the VIX for all options). Use Black-Scholes formula to price the options. Take the current term structure and linearly interpolate to find the corresponding rates. Use 360 days/year for the term structure and 250 days/year for the time to maturity of the options.

One risk driver and univariate Gaussian model [10 points]

1. Compute the daily log-returns of the underlying stock.
2. Assume they are iid normally distributed.
3. Generate 10 000 scenarios for the one-week ahead (five days) underlying price using the normal distribution fitted to the past invariants.
4. Determine the P&L distribution of the book of options, using the simulated underlying values. Assume the implied volatility stays the same. Take interpolated rates for the term structure.
5. Compute the VaR95 and the ES95.

Note. Do not forget that you are moving forward one week (five days). The maturity of the options and the rate for the term structure are therefore different than for the initial book.

Two risk drivers and bivariate Gaussian model [15 points]

1. Compute the daily log-returns of the underlying stock.
2. Compute the daily log-returns of the VIX.
3. Assume they are invariants normally distributed.
4. Generate 10 000 scenarios for the one-week ahead underlying price and the one week ahead VIX value using the normal distribution fitted to the past risk drivers.
5. Determine the P&L distribution of the book of options, using the simulated values. Take interpolated rates for the term structure.

Two risk drivers and copula-marginal model (Student-t and Gaussian copula) [20 points]

1. Proceed as before, but assume now that the first invariant is generated using a Student-t distribution with $\nu = 10$ degrees of freedom and the second invariant is generate using a Student-t distribution with $\nu = 5$ degrees of freedom. Assume the normal copula to merge the marginals. Recompute the P&L and the risk figures.

Volatility surface [20 points]

1. Fit a volatility surface to the implied volatilities observed on the market (traded call and put options). Minimize the absolute distance between the market implied volatilities and the model implied volatilities. The parametric surface is given by:

$$\sigma(m, \tau) = \alpha_1 + \alpha_2(m-1)^2 + \alpha_3(m-1)^3 + \alpha_4\sqrt{\tau}\,,$$

where $m = K/S$ is the moneyness and τ is the time to maturity of the option in years.

2. Re-price the portfolio using the parametric model. Once you have fitted the volatility surface, consider $\tilde{\sigma}(m, \tau) = \sigma(m, \tau) - (\alpha_1 + \alpha_4) + \mathrm{VIX}_T$, where VIX_T is the VIX at time T. Use this for the pricing at time T. For the pricing at $T+5$, consider $\tilde{\sigma}(m, \tau) = \sigma(m, \tau) - (\alpha_1 + \alpha_4) + \mathrm{VIX}_{T+5}$ where VIX_{T+5} is obtained by simulation.